

What a Stale Value Still Does: Mention Structure and Order Shape Score Sensitivity After In-Context Updates

Anonymous authors

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Abstract

When a context updates a fact, does the earlier value still shape a model’s answer scores? We replace a historical value in two-entity histories, keeping current assignments fixed, and measure how much more this shifts candidate scores for a query about the associated entity than about the other entity. Across six prompt constructions crossed with mention order, in seven models from five families, this sensitivity depends more on where and how the value is mentioned than on whether it was ever assigned to the queried attribute. Reversing mention order changes it by 4.970 to 11.837 nats for never-assigned values, while the contrast between a superseded and an other-attribute assignment changes it by -1.305 to 1.162 nats. A value the entity mentions in a note outweighs a superseded assignment in six of the seven models, and by more than relation status in five; in Qwen3-8B and Gemma 3 4B, with the marker **Previously** and the distance between name and value matched, the difference is 1.293 and 0.367 nats. Never-assigned values are credited to whichever entity shares their position, and adding the prefix **Previously**, lowers sensitivity in the tested constructions, but how order and the marker interact with construction differs between models. All of these are scores for a fixed set of candidate answer prefixes, not generated answers, from models with near-ceiling current-candidate ranking; in an exploratory model with lower accuracy, generated answers show a similar qualitative pattern.

1 Introduction

Conversations and documents often update a fact while leaving its earlier version in view. A user’s preference changes, an object moves, a task variable receives a new value. A language model can answer with the current value and still be influenced by the old one: its scores for candidate answers may shift when the old value changes, even though the top answer does not. This paper asks what governs that residual sensitivity.

Consider a history from our main experiment:

Previously, Nora’s badge was amber.
 Previously, Liam’s badge was coral.
 Currently, Nora’s badge is jade.
 Currently, Liam’s badge is pearl.
 What is Nora’s current badge?
 Respond with only the value, with no explanation.

The correct answer is **jade**. Suppose we replace the earlier **amber** with **violet** and leave the other prompt content fixed. The edit can shift the model’s scores for **violet** relative to **amber**. If the shift is larger when the question asks about Nora than when it asks about Liam, the score change is query-specific. We call the difference between these two shifts *query relevance*, R (Section 3.3).

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A positive R for the history above could arise for several reasons. The model may represent that **amber** was Nora’s badge, a superseded assignment to the queried attribute. It may associate **amber** with Nora because they appear in the same sentence, or link them by position because both come first in their blocks. The explicit marker **Previously** could also affect the score contrast. We compare prompt constructions that hold some features fixed while changing others, and cross them with historical and current entity order. The contrasts constrain possible explanations but do not independently identify each factor.

In seven models from five families, **query-specific score sensitivity to a historical value depends more on where and how the value is mentioned than on whether it was ever assigned to the queried attribute**. Reversing mention order changes R for a never-assigned value by 4.970 to 11.837 nats in every model, more than relation status changes it in any model, which ranges from -1.305 nats in Mistral-7B to 1.162 in Qwen3-14B. Mentioning the value with the entity instead of as a superseded assignment raises R in six of the seven models; in Mistral-7B the difference is unresolved and its sign depends on mention order. That gap exceeds relation status in five models, not resolvably in Granite-3.1-8B and Mistral-7B (Table 2). Relation status here is the contrast between superseded and other-attribute assignment, which changes both the attribute and whether the current block contradicts the earlier value. These comparisons do not rank all determinants of sensitivity; they compare the tested contrasts.

In the two models we study most closely, Qwen3-8B and Gemma 3 4B, relation status changes R by 0.314 and 0.004 nats, and an entity mention exceeds a superseded assignment by 3.504 and 2.589 nats, or 1.293 and 0.367 with the marker and name-value distance matched. Values assigned to no one are credited to whichever entity occupies the same relative position, adding the prefix **Previously**, lowers R , and finer order and marker effects differ between the two models. The two models narrowly excluded by the competence screen show the same ordering.

Our contributions are:

1. A counterfactual measure of query-specific sensitivity to a historical value, applied to six prompt constructions crossed with independently varied historical and current mention order (Sections 3–4).
2. A comparison of the prompt features that change this sensitivity: mention order produces larger changes than the relation-status contrast in every screened model; the entity-mention gap exceeds that contrast in five of seven; finer effects depend on the model (Section 5.1, Figure 1).
3. Order-stratified evidence that never-assigned values are credited by position, with effects that cancel when orders are averaged (Section 5.3).
4. A second main experiment on the prefix **Previously**, (96 histories); three controlled follow-ups, on name-value distance, on whether the other attribute is also updated, and on update chains (a development run and a score-only run); an extension of Experiment 1 to five additional screened models, giving seven models from five families; and an exploratory generation study in OLMo-2. In Qwen a separate update-control design finds a positive attribute effect and a negative update effect; in a development run of update chains every generated answer was correct, and in an independent score-only run Qwen’s sensitivity to the immediately superseded value grows with chain depth (Sections 5.4–5.9).

2 Related work

Contextual updates and knowledge conflicts. Entity-tracking benchmarks test whether models recover an entity’s state after a sequence of operations (Kim & Schuster, 2023; Tang et al., 2026), and belief-revision and in-context forgetting studies test whether models revise conclusions or stop using information once new evidence or instructions arrive (Wilie et al., 2024; Qian et al., 2026). Wang & Sun (2025) show that earlier key-value updates interfere with retrieving the latest value as the number of updates grows, and Guo et al. (2026) characterize stale-answer failures after context updates, including whether the current value remains recoverable and how attention drifts during answer selection. Contradictions between earlier and later statements concern conflicting contextual information, falling under the inter-context category of

Xu et al. (2024). Yang et al. (2026) study conflicting evidence within a single context. Most of these benchmarks emphasize answer accuracy; Guo et al. (2026) also probe recoverability and attention, and Yang et al. (2026) analyze evidence attribution. Our estimand is different: the query-specific change in candidate scores caused by a matched edit to an earlier value, measured in models selected for near-ceiling candidate-ranking accuracy, and how it depends on the way that value is mentioned.

Position in context. Where information sits in the context affects how models use it. Liu et al. (2024) show that retrieval accuracy depends strongly on the position of the relevant passage in long inputs, and Yang et al. (2026) identify a positional preference as an obstacle to resolving conflicts. Our contexts are six lines long, far from the long-context regime, yet relative position still determines which entity a never-assigned value is credited to.

Entity binding. Mechanistic studies analyze how models bind entities to attributes in context and retrieve those bindings, including binding-ID representations (Feng & Steinhardt, 2024), mixtures of positional, lexical, and reflexive retrieval mechanisms (Gur-Arieh et al., 2026), and lookback mechanisms for tracking characters’ beliefs (Prakash et al., 2026). These works intervene on internal activations. We work at the level of prompts and output scores, which lets us compare how different surface constructions feed whatever binding mechanism the model uses, but not identify the mechanism itself.

Persistence after knowledge editing. Knowledge-editing studies ask what remains of a fact after it is edited, whether by changing parameters or through an external memory: original facts stay linearly decodable after edits by weight-changing editors and by GRACE, which leaves the weights unchanged and overrides the fact through an external memory (Srivastava & Chatterjee, 2026), and they can resurface after superficially successful edits (Xie et al., 2025). In our setting both the earlier and the current value are present in the input and the model is not edited; we measure sensitivity to an earlier value that is still in view.

3 Task and measurement

3.1 Histories and prompt constructions

Each item, which we call a *history*, specifies two entities (for example Nora and Liam), a target attribute (badge, color, code, or label), and six distinct values drawn from an eight-value vocabulary: `amber`, `coral`, `jade`, `pearl`, `slate`, `teal`, `violet`, `ivory`. Two values are the entities’ earlier values, two are their current values, and two serve as replacement (donor) values for the edits. For analysis we label the entities x and z ; which name plays which role is counterbalanced.

A history is rendered in six *constructions* that differ only in how the two earlier values enter the context (Table 1). Every construction except the live control contains the current-state block C (Currently, Nora’s badge is jade. and Currently, Liam’s badge is pearl.), and every prompt ends with What is Nora’s current badge? (or the Liam query) and Respond with only the value, with no explanation. Each sentence is on its own line.

In the live control the edited value is the current answer, so the edit changes the correct answer for the query about the edited entity; it calibrates the size of a fully relevant edit. In the other five constructions both current values stay fixed, so the correct answer never changes.

Two order factors are crossed with every construction. *Historical order* sets whether x ’s or z ’s earlier value is mentioned first; *current order* independently sets whether x ’s or z ’s current assignment comes first. We call the orders *aligned* when both blocks put the same entity first and *reversed* otherwise. For the unassigned constructions, the earlier values carry the labels x and z only for the analysis; the text assigns them to no one, so aligned order means that the first unassigned value appears in the same ordinal slot as the entity listed first in C . The live control has a single assignment block, ordered by the historical-order factor.

Table 1: The six constructions, shown for a history in which Nora’s and Liam’s earlier values are **amber** and **coral** and their current values are **jade** and **pearl**. C is the current-state block. In the other-attribute construction the earlier values belong to **team** or **project** (balanced across histories).

Construction	Context	Earlier value is...
Superseded	Previously, Nora’s badge was amber . Previously, Liam’s badge was coral . C	assigned to the queried attribute, then updated
Other attribute	Previously, Nora’s team was amber . Previously, Liam’s team was coral . C	assigned to a different attribute
Entity mention	Nora mentioned amber in an unrelated note. Liam mentioned coral in an unrelated note. C	mentioned with the entity, never assigned
Early unassigned	The unassigned badge value was amber . The unassigned badge value was coral . C	mentioned without an entity, before C
Late unassigned	C The unassigned badge value was amber . The unassigned badge value was coral . C	mentioned without an entity, after C
Live	Currently, Nora’s badge is amber . Currently, Liam’s badge is coral .	the current value (positive control)

3.2 Bounded answer-prefix score

Let p be a tokenized prompt, including the model’s chat template and the assistant prefix **Answer:.** For a token continuation $w = (w_1, \dots, w_m)$,

$$L(w \mid p) = \sum_{t=1}^m \log P(w_t \mid p, w_{<t}). \quad (1)$$

Each value v has a set $\mathcal{W}(v)$ of tokenizer-validated surface forms: lowercase and title case, each with and without one leading space. Validation keeps only forms whose tokens leave the prompt’s tokenization unchanged, removes duplicate token sequences, and rejects any set in which one candidate’s tokens are a prefix of another’s, so the retained events are disjoint. The score of v is their total log probability mass,

$$S(v \mid p) = \log \sum_{w \in \mathcal{W}(v)} \exp L(w \mid p), \quad (2)$$

without length normalization or an end-of-answer token. S is therefore the log probability mass of a fixed, finite set of candidate answer prefixes, which is what we mean by a *bounded* score; it is not the probability of a complete generated answer. Candidate-ranking accuracy compares S across the eight vocabulary values.

3.3 Matched edits and query relevance

An edit replaces one occurrence of a source value s with a donor value r , leaving every other token unchanged. Let p^0 and p^1 be the baseline and edited prompts for history h , construction c , edited entity $e \in \{x, z\}$, query q , and order cell o . The edit effect is the change in the donor-versus-source score difference:

$$E_{h,c,e,q,o} = [S(r \mid p^1) - S(s \mid p^1)] - [S(r \mid p^0) - S(s \mid p^0)]. \quad (3)$$

A positive E means that the edit moves the scores toward the donor. In the example of Section 1, replacing Nora’s earlier **amber** with **violet** can raise **violet** relative to **amber** even though **jade** remains the answer.

Write the order cell as $o = (o_H, o_C)$, with each component 0 for x first and 1 for z first. For an order group g , let $\bar{E}^g$ average E over its cells: all four cells (overall), (0,0) and (1,1) (aligned), or (0,1) and (1,0) (reversed). With q_x and q_z the queries about x and z , *query relevance* is

$$R_{h,c}^g = \frac{1}{2} [\bar{E}_{h,c,x,q_x}^g - \bar{E}_{h,c,x,q_z}^g + \bar{E}_{h,c,z,q_z}^g - \bar{E}_{h,c,z,q_x}^g]. \quad (4)$$

R is the extra shift under the query about the edited entity, compared with the query about the other entity, averaged over edits to both entities. It is zero when the edit moves scores equally for both queries, however large that common shift is. Because the contrast is symmetric, additive effects of the edited label and of the query cancel; interactions between them, including order-dependent ones, do not, which is why we report aligned and reversed estimates as well as the overall mean. We omit g for the overall estimate.

3.4 What each comparison changes

Some prompt features change together by design and cannot be separated. *Superseded versus other attribute* is the comparison for relation status. The two share the sentence template, the marker **Previously**, the position, and the entity–value pairing, but a superseded value was assigned to the queried attribute and is contradicted by the current block, while an other-attribute value was assigned to an attribute the current block never mentions; the contrast measures both differences at once, and a follow-up separates them (Section 5.5). *Entity mention versus superseded* is the comparison for construction: both pair each value with its entity, but the superseded line assigns the value and carries **Previously**, while the entity-mention line only mentions it, with a different predicate and word order. *Aligned versus reversed order* is the comparison for position; in the unassigned constructions it is the only link between a value and an entity. The marker is varied separately in Experiment 2 (Section 4.3); the marker-absent superseded sentence remains in the past tense. All of these are comparisons between prompts; they do not identify the internal computation that produces the scores.

4 Experimental setup

4.1 Models and competence screen

We study two models most closely, the instruction-tuned Qwen3-8B (Yang et al., 2025) and Gemma 3 4B (Gemma Team, 2025), loaded with int8 weights (Dettmers et al., 2022) and float16 (Qwen) or bfloat16 (Gemma) computation, using each model’s chat template with thinking disabled for Qwen3; without a size, *Qwen* and *Gemma* refer to these two. Revisions and software are listed in Appendix C. A model enters the screened analysis after passing a competence screen on a separate set of 24 histories. The screen selects models with near-ceiling candidate-ranking accuracy; confirmation-set analyses retain all histories, including ranking errors. In every construction, the correct current value must be strictly ranked first among the eight candidates on at least 99% of the distinct screening prompts (ties count as errors), and in the superseded construction the mean margin of the current over the earlier value must be positive. Both models passed, with lowest construction-level accuracies of 287/288 (Qwen, live) and 575/576 (Gemma, entity mention). Seven further models were screened in the same way and the five that passed were run through Experiment 1 (Section 5.9): Falcon3-7B-Instruct (Falcon-LLM Team, 2024), Granite-3.1-8B-Instruct (IBM Granite Team, 2024b;a), Mistral-7B-Instruct-v0.3, Gemma 3 12B and Qwen3-14B. OLMo-2-7B-Instruct (Team OLMo, 2025) and Qwen2.5-7B-Instruct (Qwen Team, 2024) failed, as did Phi-4-mini (Abouelenin et al., 2025); Llama-3.1-8B-Instruct was unavailable because access to its gated weights was declined. We evaluate Mistral-7B-Instruct-v0.3, an instruction-tuned checkpoint from the Mistral 7B family (Jiang et al., 2023; Mistral AI, 2024). We cite the Granite 3.0 technical report for family background (IBM Granite Team, 2024a) and the Granite 3.1 model card for the evaluated checkpoint (IBM Granite Team, 2024b).

4.2 Analysis and prespecification

All estimates are computed per history and then averaged over histories; histories are the unit of uncertainty. Intervals are percentile 95% intervals from 2,000 bootstrap resamples of histories (seed 73021). Models are scored on the same histories and reported separately, without pooling. The intervals quantify variation across constructed histories, with names, templates, vocabularies, and models fixed or narrowly sampled; intervals for exploratory analyses are pointwise, not simultaneous.

The design, competence thresholds, and analysis of Experiment 1 were fixed in the project repository before its confirmation data were scored, and a dated amendment recorded before the results were examined specified the decision rule for the original hypothesis (Section 5.2). The 24-history run of Experiment 2 was likewise

specified before scoring. The 96-history run of Experiment 2 and the follow-up experiments were designed and their design documents hash-stamped on the experiment machine before any data were generated, but were not committed to version control. No experiment was externally preregistered. The reversed-order versions of the three primary contrasts were prespecified as part of that decision rule; all other order-stratified estimates, paired aligned-minus-reversed differences, per-attribute strata, and analyses restricted to correctly answered histories are exploratory.

4.3 Experiments

Experiment 1: constructions crossed with order. 96 histories: each of six entity pairs is crossed with the four target attributes and both assignments of names to the x and z roles, with two histories per cell; the other-attribute construction uses **team** in half of them and **project** in the other half. Each history yields $6 \times 2^5 = 192$ scored prompts (construction, query, edited entity, historical order, current order, baseline versus edit), 18,432 per model. Tokenizer-aware audits verified that each baseline/edit pair differs only at the intended one-token span. An earlier, simpler comparison that motivated this design, with a single unassigned control, is reported in Appendix A.

Experiment 2: the explicit marker. In Experiment 1 the superseded and other-attribute constructions carry **Previously** and the others do not. Experiment 2 crosses construction (superseded, entity mention) with marker (absent, present), where adding the marker changes nothing but the prefix **Previously**, on each historical line (Table D1), with the other factors crossed as in Experiment 1. It uses 96 histories with one entity pair and the attribute **badge**, so its R values are compared within the experiment; an earlier 24-history run of the same design is reported in Appendix D. The prespecified quantities are the *marker effect*, $R_{\text{present}} - R_{\text{absent}}$, within each construction, and the *interaction*, the superseded minus the entity-mention marker effect.

Follow-ups and extensions. Beyond the two main experiments, three controlled follow-ups address questions they leave open, each with 96 histories, the same scoring and analysis, and a recorded prediction (designs in Appendix F); the additional models of Section 4.1 extend Experiment 1, and one study is exploratory. *Updated other attribute* adds a construction in which the other attribute is also updated, varying whether the earlier value was the queried attribute and whether it was contradicted by an update; the discount account predicts a negative update effect. *Name-value distance* crosses construction with a near or far name-value distance, measured in tokens before data generation, with **Previously** on every line; the prediction was near above far in both constructions. *Update chains* give the queried attribute three to five successive assignments, in a development run that looked for a depth at which answers fall below ceiling and a separate score-only run that estimates how R changes with depth. An exploratory study of *generated answers* uses OLMo-2-7B-Instruct, which fails the screen and answers below ceiling; its generated answers are summarized by *donor following*, a descriptive measure on edited prompts only (Appendix F).

5 Results

5.1 What changes query relevance

Figure 1 sets the comparisons of Section 3.4 side by side for Qwen and Gemma. Relation status matters little on average: superseded relevance exceeds other-attribute relevance by 0.314 nats (95% CI [0.147, 0.482]) in Qwen and by 0.004 [−0.104, 0.118] in Gemma. The figure adds two comparisons from a separate update design (Section 5.5). Construction matters more, with entity-mention relevance above superseded relevance by 3.504 [3.198, 3.806] and 2.589 [2.361, 2.829] nats, and a smaller gap remains with the marker and name-value distance matched (Section 5.6). The rows of Figure 1 are comparisons from different designs and stimulus sets, not a factorial decomposition, so they do not add up to one another. Mention order decides the sign for never-assigned values. Reversing the order lowers early-unassigned relevance by 6.578 [6.356, 6.789] nats in Qwen and 4.970 [4.700, 5.257] in Gemma, taking it from positive to negative, while the same reversal moves superseded relevance by only 0.340 [0.218, 0.470] and 0.032 [−0.101, 0.168]. For entity mentions the aligned-minus-reversed difference has opposite signs in the two models: −1.969 [−2.344, −1.604] in Qwen

and +1.810 [1.580, 2.058] in Gemma. In Experiment 2, adding the prefix **Previously**, lowers relevance in the tested constructions by between 0.652 and 1.734 nats in each construction and model.



Figure 1: What changes query relevance. Every row is a difference in R (nats) between two prompt variants, with 95% history-bootstrap intervals. Relation-status comparison: superseded minus other attribute (Experiment 1), then two comparisons from the separate updated-other-attribute design: superseded minus updated other attribute, and updated minus static other attribute. Construction comparisons: entity mention minus superseded (Experiment 1), then the same gap with the marker and name-value distance matched (distance follow-up, near condition; Table 3). Order comparisons: R under aligned minus R under reversed order, paired within history. Marker comparisons: R with minus without the prefix “Previously”, within each construction. The grey tag at the right of each row names its design; rows from different designs use different prompts and are not a factorial decomposition. Every design uses 96 histories, and Experiment 2 uses one entity pair and one attribute. Estimates are computed from the sealed analyses by `scripts/fig_decomposition.py`.

Table 2 gives the main comparisons for all seven models that passed the competence screen. In every one, reversing mention order moves early-unassigned relevance from positive to negative, by more than relation status moves relevance in any model. Entity mentions outweigh superseded assignments in all but Mistral-7B, and the paired difference between that gap and relation status is resolved in all but Granite-3.1-8B and Mistral-7B (Section 5.9). R for every construction and order group, including the live control (32.598 nats in Qwen, 39.355 in Gemma), is in Appendix B.

These score changes coexist with correct rankings. In the superseded construction both models ranked the current value strictly first in all 1,536 baseline and all 1,536 edited prompts, with mean current-minus-earlier margins of 21.734 nats (Qwen) and 23.854 (Gemma). The edited source and donor candidates both had low

Table 2: Experiment 1 in every model that passed the competence screen: (a) the relation-status contrast, (b) the entity-mention gap, early-unassigned relevance by order group, and (b) minus (a) paired within history. Means with 95% history-bootstrap intervals over 96 histories, in nats; models are reported separately.

Construction contrasts			
Model	(a) Superseded	(b) Entity mention	(b) - (a)
	- other attribute	- superseded	
Qwen3-8B	0.314 [0.147, 0.482]	3.504 [3.198, 3.806]	3.191 [2.811, 3.567]
Gemma 3 4B	0.004 [-0.104, 0.118]	2.589 [2.361, 2.829]	2.585 [2.317, 2.861]
Falcon3-7B	-0.052 [-0.168, 0.058]	1.203 [0.928, 1.494]	1.254 [0.929, 1.596]
Granite-3.1-8B	1.001 [0.880, 1.125]	1.372 [1.015, 1.741]	0.371 [-0.019, 0.744]
Mistral-7B	-1.305 [-2.048, -0.648]	-0.415 [-1.000, 0.187]	0.889 [-0.134, 1.986]
Gemma 3 12B	0.442 [0.249, 0.631]	1.321 [0.987, 1.645]	0.879 [0.423, 1.351]
Qwen3-14B	1.162 [0.988, 1.351]	2.380 [2.045, 2.698]	1.218 [0.770, 1.640]

Positional contrasts		
Model	Early unassigned: aligned	Early unassigned: reversed
Qwen3-8B	3.252 [3.046, 3.455]	-3.327 [-3.518, -3.150]
Gemma 3 4B	2.598 [2.395, 2.798]	-2.372 [-2.567, -2.178]
Falcon3-7B	3.866 [3.699, 4.038]	-3.893 [-4.082, -3.695]
Granite-3.1-8B	5.105 [4.761, 5.477]	-5.616 [-5.920, -5.309]
Mistral-7B	5.975 [5.478, 6.481]	-5.862 [-6.460, -5.309]
Gemma 3 12B	3.772 [3.533, 4.027]	-3.316 [-3.571, -3.063]
Qwen3-14B	3.686 [3.391, 3.978]	-3.987 [-4.296, -3.667]

absolute log masses: in Qwen’s superseded construction the source fell from -22.271 to -23.881 nats on average and the donor rose from -23.618 to -22.120 , against a current candidate near zero (Table B8), so their changes do not imply that probability moves from one to the other, or a matching change in generated answers.

5.2 Relation status and construction

Superseded minus other attribute is the study’s closest estimate of what being an obsolete value of the queried attribute adds (Section 3.4 explains why it joins two properties). In Qwen the 0.314-nat average, about a sixth of superseded relevance, masks heterogeneity across target attributes, from -0.570 for `code` to 0.876 for `color` (Table B11); Gemma’s attribute-specific intervals all include zero, which is not evidence of equivalence. Mentioning the earlier value in a note about the entity, without assigning it, produces more relevance than the superseded assignment in both models and for every target attribute (Table B10). These contrasts, with superseded minus early unassigned (Table B2), were the test of our original hypothesis, which predicted that superseded relevance would exceed all three controls, each with a lower 95% bound above zero, both overall and under reversed order. The prediction fails in both models, because the entity-mention contrast is negative overall and under reversed order (Table B4).

5.3 Mention order

When no sentence links a value to an entity, the models link them by position. Under aligned order the first unassigned value sits in the same slot as the entity listed first in the current block, and an edit to it is more relevant to that entity’s query; under reversed order the same value lines up with the other entity, and relevance turns negative. The pattern holds for early and late unassigned mentions in both models, with paired order effects between 4.970 and 6.639 nats and positive in every one of the 96 histories (Table B3). We say that such a value is *credited* to an entity as shorthand for this query-specific score contrast, which follows matching ordinal positions across the historical and current blocks. It is distinct from primacy or recency: the unassigned lines occupy the same positions in both order groups, and early and late placement show the same sign pattern. This behavior is consistent with slot-based matching of values to entities in

these prompts. Related mechanistic work identifies abstract binding IDs (Feng & Steinhardt, 2024) and ordering IDs that track entities by their order of introduction (Prakash et al., 2026). None of this identifies the mechanism here.

Averaging over orders hides this pattern: overall unassigned relevance is near zero (-0.037 and 0.001 in Qwen; 0.113 and 0.125 in Gemma) not because the edit is ignored (an early unassigned edit shifts scores by 6.234 to 8.239 nats under either query in both models) but because aligned and reversed effects cancel. Explicit association keeps relevance positive in both order groups: superseded relevance changes little with order, while entity-mention relevance changes in opposite directions in the two models (Figure 3, Appendix B).

5.4 The explicit marker

Experiment 2 adds or removes the prefix **Previously**, with the rest of each prompt fixed (Table D1). Adding the prefix **Previously**, lowers R in the tested constructions in both models, with every within-construction interval below zero (Table D2). In Qwen the reduction is larger for entity mentions: the marker-by-construction interaction is 1.082 [$0.850, 1.308$], positive in both order groups. In Gemma the pooled interaction, 0.125 [$-0.012, 0.265$], is unresolved because the two order groups have opposing, resolved interactions: 0.482 [$0.307, 0.655$] under aligned and -0.232 [$-0.436, -0.027$] under reversed order, a paired aligned-minus-reversed difference of 0.715 [$0.461, 0.968$] (in Qwen -0.896 [$-1.191, -0.577$]; Figure 4). The earlier 24-history run also found negative marker effects, a positive pooled interaction in Qwen, and an unresolved pooled interaction in Gemma; its order-specific Gemma intervals include zero, so it did not establish the crossover (Appendix D). Relevance stays positive with the marker in every construction and order group, so adding the prefix lowers query-specific sensitivity to the earlier value without removing it. Without the marker, entity-mention relevance still exceeds superseded relevance, by 3.059 [$2.758, 3.370$] nats in Qwen and 0.908 [$0.783, 1.034$] in Gemma, although in Gemma’s reversed-order cell the difference is not resolved (-0.175 [$-0.357, 0.011$]). An exploratory Phi-4-mini run showed a third pattern (Appendix D).

5.5 Separating relation status

The updated-other-attribute follow-up varies whether the other attribute is also updated. With prompt length and layout matched, having been the queried attribute raised relevance (superseded minus updated other attribute: 0.740 [$0.661, 0.818$] nats in Qwen, 0.148 [$0.038, 0.261$] in Gemma), while being contradicted by an update lowered it in Qwen (updated minus static other attribute: -0.289 [$-0.363, -0.215$]) but not detectably in Gemma (-0.028 [$-0.137, 0.076$]; Table F4). In this design Gemma ranks the current candidate first on only 2,895 of 3,072 superseded prompts, below the 99% screening level (Table F10). Restricting to histories in which every prompt is ranked correctly preserves the positive queried-attribute effects and Qwen’s negative update effect; Gemma’s update effect remains unresolved: 0.734 [$0.662, 0.811$] and -0.276 [$-0.353, -0.200$] in Qwen (87 histories), 0.277 [$0.080, 0.481$] and an unresolved 0.047 [$-0.123, 0.221$] in Gemma, although only 28 of Gemma’s 96 histories qualify. Both contrasts are conditional: the first compares queried-attribute status when both assignments are updated, the second compares updating when the assignment concerns another attribute. In Qwen the result supports an account in which a positive attribute effect and a negative update effect partly cancel; the follow-up uses different prompts and does not decompose the Experiment 1 estimate. The recorded prediction of the discount account holds for Qwen and is unresolved for Gemma.

5.6 Why entity mentions outweigh assignments

Matching the name-value token gap does not remove the construction gap in the tested templates. Moving the name next to the value raised superseded relevance in both models (near minus far: 0.500 [$0.400, 0.599$] in Qwen, 0.393 [$0.288, 0.496$] in Gemma) but not entity-mention relevance (-0.370 [$-0.498, -0.249$] and -0.027 [$-0.110, 0.058$]; Table F1), so the recorded prediction, near above far in both constructions, fails in both models. Table 3 lists the gap in each tested combination of marker placement and name-value distance; it is smallest, 1.293 and 0.367 nats, with the marker on both lines and a one-token name-value distance in both. The rows come from different stimulus sets, so they are matched comparisons rather than an additive decomposition, and the distance rewrites also change syntax; the far conditions are not matched either

Table 3: Entity-mention minus superseded relevance (nats) in each tested combination of marker placement and name-value distance, with 95% history-bootstrap intervals over 96 histories. “Marker” gives which line carries **Previously**; “Name-value gap” gives the measured tokens between name and value (entity mention vs. superseded). Rows within an experiment share stimuli; rows from different experiments do not (Experiment 2 uses one entity pair and one attribute).

Experiment	Marker	Name-value gap	Qwen3-8B	Gemma 3 4B
Experiment 1	superseded only	1 vs. 3-4	3.504 [3.198, 3.806]	2.589 [2.361, 2.829]
Experiment 2	neither	1 vs. 3-4	3.059 [2.758, 3.370]	0.908 [0.783, 1.034]
	both	1 vs. 3-4	1.976 [1.795, 2.169]	0.782 [0.673, 0.894]
Distance follow-up	both	7 vs. 3-4 (far)	2.163 [1.866, 2.459]	0.788 [0.657, 0.922]
	both	1 vs. 1 (near)	1.293 [1.019, 1.571]	0.367 [0.234, 0.514]

(seven tokens for entity mention against three or four for superseded), so this is not a factorial distance manipulation. Predicate and framing are the remaining candidate explanations, alongside other wording differences.

5.7 Below ceiling

In the exploratory study of OLMo-2-7B-Instruct, which fails the screen and answers below ceiling, generated answers were correct for 2,950 of 3,072 superseded prompts but only 1,846 of 3,072 entity-mention prompts and 1,528 and 2,131 of the early- and late-unassigned prompts (Table F5); most errors name one of the two earlier values or no valid value (Table F6). Donor following, a separate descriptive measure on edited prompts only, shows that editing a value mentioned with the queried entity moved answers to the donor query-specifically (0.155 [0.124, 0.190]); never-assigned values pulled answers toward the entity in the matching position (early unassigned: 0.161 [0.122, 0.203] aligned, -0.164 [-0.214 , -0.120] reversed); superseded values almost never surfaced, although the superseded edit still moved answers by a small, query-specific amount (0.008 [0.003, 0.014]). A baseline-adjusted generation contrast gives the same ordering, 0.320 [0.268, 0.379] for entity mention against 0.017 [0.007, 0.029] for superseded (Table F7).

OLMo-2 is separate behavioral evidence from below ceiling. Its generated answers follow the construction and position ordering of most screened models’ scores: mentioned and position-aligned values pull answers toward the donor, and superseded values rarely do. Its own candidate-score R does not: 6.157 [5.933, 6.393] nats for superseded against 5.350 [5.040, 5.669] for entity mention. In this model, then, R does not predict generated-answer behavior, and OLMo-2’s candidate-score ordering differs from its generated-answer ordering.

5.8 Depth of updates

Chain depth is the total number of successive assignments of the queried attribute, including the current one, so depth 3 means two updates. Two separate runs address depth. In a development run, both models generated the correct answer for every prompt at depths 3, 4, and 5, so by the recorded rule no generation confirmation was run (Appendix F). An independent score-only run on 96 fresh histories per depth then estimated how R changes with depth; it collected no generated answers, and its current-candidate rank accuracy, which is not generated-answer accuracy, was at least 3,067 of 3,072 at every depth in both models (Table F10). In that run Qwen’s R for the edited immediately superseded value rose from 7.353 [6.972, 7.725] nats at depth 3 to 10.762 [10.303, 11.220] at depth 4 and 11.835 [11.424, 12.286] at depth 5; the depth-5-minus-depth-3 difference of 4.482 [3.915, 5.047] meets the recorded criterion. Gemma’s R stayed between 3.450 and 3.708 (depth 5 minus depth 3: 0.258 [-0.063 , 0.557]). Deeper chains are also longer and add preceding text, so the growth need not come from the number of updates as such; the interference that Wang & Sun (2025) observe in answers as updates accumulate is an analogy, not an identified cause.

5.9 Additional models

Of seven additional models, five passed the competence screen: Falcon3-7B, Granite-3.1-8B, Mistral-7B, Gemma 3 12B and Qwen3-14B; OLMo-2-7B-Instruct and Qwen2.5-7B-Instruct failed (Appendix F). In Table 2, relation status ranges from -1.305 $[-2.048, -0.648]$ nats in Mistral-7B through no resolvable effect in Falcon3-7B to 1.162 $[0.988, 1.351]$ in Qwen3-14B, about a quarter of its superseded relevance. Within each family the larger model weights relation status more and the entity-mention gap less (Qwen3-8B to 14B: 0.314 to 1.162 and 3.504 to 2.380 ; Gemma 3 4B to 12B: 0.004 to 0.442 and 2.589 to 1.321); with two sizes per family this is an observation, not a trend. In Granite-3.1-8B and Mistral-7B the entity-mention gap does not resolvably exceed relation status (paired differences 0.371 $[-0.019, 0.744]$ and 0.889 $[-0.134, 1.986]$), and these two models have the largest positional effects. Mistral-7B’s entity-mention gap changes sign with mention order (Appendix F). Although it passed the screen, Mistral-7B ranks the current candidate first on only 3,020 of 3,072 other-attribute confirmation prompts (98.3%; Table F10). On the 68 histories in which every prompt is ranked correctly, its relation-status contrast is -0.037 $[-0.680, 0.541]$, while its unresolved entity-mention gap and its positional effect remain; its negative relation status therefore depends on histories with ranking errors.

The threshold excludes two models narrowly, Phi-4-mini and Qwen2.5-7B, each below 99% in one construction. Scored descriptively, both meet the two criteria recorded before scoring: the entity-mention gap exceeds the relation-status contrast in absolute value, with (b) $-$ (a) and (b) $+$ (a) both above zero, and early-unassigned relevance is positive under aligned and negative under reversed order (Table F9). In both, superseded relevance falls below other-attribute relevance.

6 Discussion

A model’s sensitivity to an earlier value tracks how that value is attached to an entity in the text more than what the value used to mean. A superseded value of the queried attribute and a never-updated value of another attribute differ by about a third of a nat or less in Qwen3-8B and Gemma 3 4B and by -1.305 to 1.162 nats in the other screened models, although only the first is contradicted by the current state. In six of the seven models a value placed in a sentence with the entity, even a note described as unrelated, produces more query-specific sensitivity than a superseded assignment, and a value attached to no one is credited to whichever entity shares its position. For the question that motivated this work, whether models retain obsolete assignments, the residual sensitivity is not specific to obsolescence, at least in the averages and directions the models share (Table 2). Beyond position, the ranking has exceptions. In Granite-3.1-8B the entity-mention gap and relation status are of similar size. In Mistral-7B the construction difference changes sign with mention order and superseded values count less than other-attribute values. An update discount larger than the positive queried-attribute effect could produce this pattern. Qwen’s follow-up establishes partial cancellation, not dominance of the update effect; neither explanation was tested in Mistral, and Mistral’s negative estimate becomes unresolved when histories with ranking errors are excluded. In the larger Qwen and Gemma models relation status is resolved and positive.

Within Qwen and Gemma, the directions of the finer effects also differ: reversing mention order raises entity-mention relevance in Qwen and lowers it in Gemma, and the marker-by-construction interaction is positive in Qwen in both order groups but reverses sign between order groups in Gemma. The data therefore support a claim about which prompt features carry weight, not a single rule that predicts their direction in every model. The positional result connects these short prompts to findings in longer ones: Liu et al. (2024) and Yang et al. (2026) report that where evidence appears changes how models use it. Prior work identifies abstract binding IDs (Feng & Steinhardt, 2024), ordering IDs (Prakash et al., 2026), and positional, lexical, and reflexive retrieval mechanisms (Gur-Arieh et al., 2026). Our prompt-level results do not distinguish among these mechanisms. For evaluation, averaging over mention orders can produce null effects that conceal large opposing ones.

Why entity mentions outweigh superseded assignments remains open. The gap persists without the marker and, in the tested templates, when the name-value token gap is matched, although those rewrites also change

syntax. The constructions still differ in predicate (**mentioned** versus **was**) and in whether the value is framed as an attribute at all.

Score changes on low-probability candidates expose residual sensitivity to the prompt while the current candidate stays dominant, which answer accuracy cannot show. The matched controls then test whether that sensitivity is specific to obsolete assignments. Magnitudes in log score do not translate into practical error rates.

Limitations. First, the competence screen is an admission criterion on a separate set of 24 histories; confirmation-set accuracy is reported separately and falls below the screening level for Mistral-7B (Section 5.9). Second, model size: at the sizes tested (4B to 14B parameters), position exceeds relation status in every model and construction does in five of seven, and within both families the larger model gives relation status more weight, an observation from two sizes per family, not a scaling trend; whether relation status overtakes construction or position in larger models is untested and would qualify the main claim. Third, all results concern the tested implementations with int8 weights; native precision was not evaluated. Fourth, the measure is narrow by design: scores are bounded masses of fixed answer prefixes, and the edited candidates have low absolute scores, so their changes do not imply that probability moves between them or that a generated error becomes likely. Fifth, the models are at or near ceiling in candidate ranking (Table F10). In Qwen and Gemma, neither appended distractor entities (Appendix E) nor the development run of update chains up to depth 5 produced a generated answer naming an earlier value, and the 96-history chain run scored candidates only. These generation runs did not reveal stale-value answers; whether the measured score sensitivity predicts generated-answer changes remains unestablished. OLMo-2, an exploratory model with lower accuracy, shows a similar qualitative pattern in its generated answers, but its own candidate scores order the constructions differently. The derived-code task (Appendix A.2) brings Qwen’s candidate ranking below ceiling but lacks the Experiment 1 controls and generated answers; a confirmatory test would need a screened model below ceiling, or a task with those controls that brings Qwen and Gemma there, and generated answers. Finally, names, attributes, templates, vocabularies, the language, and the models are fixed or narrowly sampled, so the intervals do not quantify generalization beyond these stimuli.

7 Conclusion

We measured how much an edit to an earlier, updated value shifts a language model’s candidate scores for a query about the associated entity, relative to a query about the other entity. In seven models from five families this sensitivity depends more on where and how the value is mentioned than on whether the value was ever assigned to the queried attribute: among the tested contrasts, mention order produces the largest changes in every model, an entity mention outweighs a superseded assignment in six of the seven, and relation status ranges from a negative effect in Mistral-7B to about a quarter of superseded relevance in Qwen3-14B. Finer order and marker effects differ between models. Because the screened models have near-ceiling current-candidate ranking, these are statements about candidate scores, not generated answers. Studies of how models handle updated information should control the construction and order of earlier mentions, and report results per model, before attributing residual sensitivity to obsolete assignments.

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A The preliminary comparison and a derived-code task

A.1 Preliminary comparison

Experiment 1 was motivated by an earlier, simpler comparison with different sentence templates. It contrasted superseded assignments with a single control in which the two earlier values were mentioned as unassigned values, averaging the two mention orders. Superseded relevance exceeded this control in all three models that passed that comparison’s competence screen: by 3.50 nats (95% CI [3.20, 3.80]) in Qwen3-8B, 1.76 [1.51, 2.03] in Gemma 3 4B, and 5.53 [5.27, 5.79] in Phi-4-mini. Read alone, this looks like evidence that superseded assignments retain query-specific influence. But the control matched neither the entity association nor the relative position of the superseded mentions, so the comparison could not say which property produced the difference. Experiment 1 adds controls for each.

The preliminary comparison used different templates from Experiments 1 and 2. A superseded prompt for the Nora/Liam example read:

The badge assigned to Nora was amber. The badge assigned to Liam was coral.
--

```

Later, Nora's badge was changed to jade.
Later, Liam's badge was changed to pearl.
What is Nora's current badge?
Respond with only the value, with no explanation.

```

The control stated the current assignments and mentioned `amber` and `coral` as unassigned badge values, in both mention orders; the primary contrast averaged the two orders within each history. A live control edited the current assignment. Scoring, E , and R were defined as in Section 3. Each history contributed 40 scored prompts, giving 3,840 prompts and 1,920 matched edit pairs over 96 confirmation histories per model, with the same history-level bootstrap. Four models were screened with the same 99% rule. Qwen3-8B, Gemma 3 4B, and Phi-4-mini passed; Mistral-7B-Instruct-v0.3 scored 281/288 on the unassigned control and was not run.

A.2 Derived-code task

To ask whether sensitivity to an earlier value reaches a computation that uses it, we mapped values to shuffled opaque codes and asked which code corresponded to an entity’s current badge. The codebook stayed fixed across baseline and edit; editing an earlier badge value changed which code it mapped to, not the codebook or the current assignment. We scored all 16 full code sequences by their unnormalized log probability. Only Qwen passed this task’s screen (48/48); its threshold, 47/48 (97.9%), was specific to this task and recorded before scoring. Gemma scored 45/48 and Phi 33/48. For Qwen, query relevance was 5.544 nats (95% CI [5.159, 5.941]) for an edit to the earlier value and 12.836 [12.418, 13.257] for an edit to the current value, both positive in all 96 histories. Qwen’s candidate ranking is below ceiling here: with the earlier value edited, the current code ranked first in 360/384 baseline and 364/384 edited prompts; 9 pairs changed from incorrect to correct and 5 from correct to incorrect. Changes in the current code’s log probability (0.108 [−0.050, 0.264]) and in the candidate margin (0.103 [−0.118, 0.325]) had intervals spanning zero. This task lacks the entity-mention and order controls of Experiment 1 and collected no generated answers, so it shows score sensitivity in a derived computation, below ceiling in candidate ranking, without saying which property of the earlier mention produces it or whether generated answers change.

B Additional results for Experiment 1

All intervals in this appendix are 95% history-bootstrap intervals (Section 4.2).

B.1 Query relevance by construction and order

Table B1 and Figure 2 give R for every construction and order group, and Table B2 the three superseded-minus-control contrasts that tested the original hypothesis. Superseded and other-attribute relevance are positive and similar in both order groups; entity-mention relevance is the largest among the non-live constructions; unassigned relevance is near zero overall but large and of opposite sign in the two order groups.

B.2 Order groups

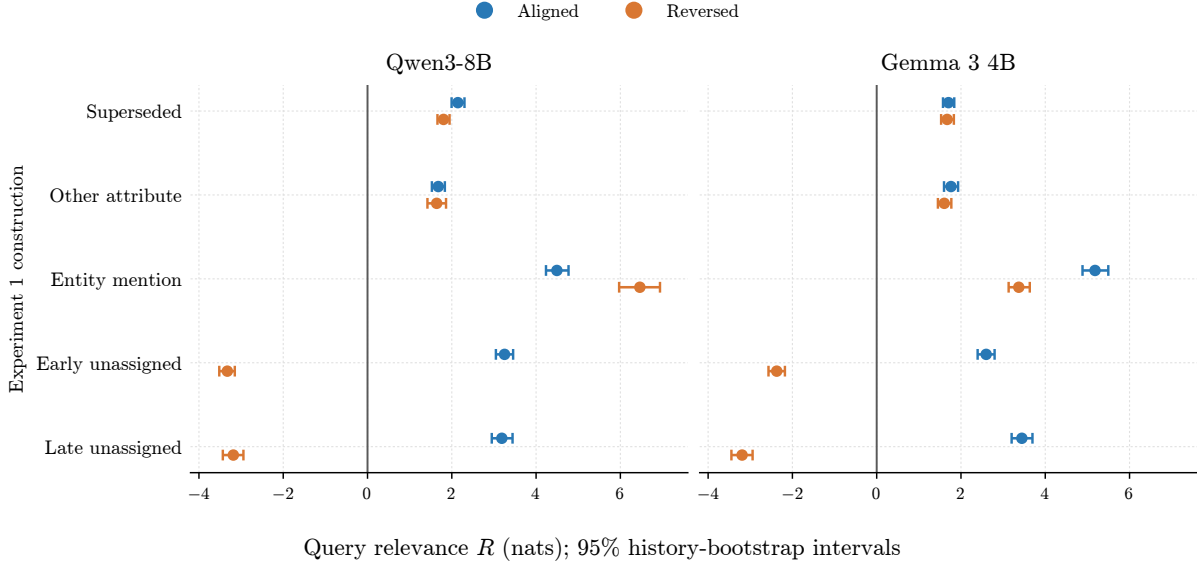
Table B3 gives the paired order effect on R itself for each construction, Table B4 each superseded-minus-control contrast by order group, and Table B5 the paired difference between the aligned and reversed contrasts. Paired differences are computed within history. The live control has one assignment block and no order split.

B.3 Accuracy and correctly answered histories

Table B6 gives strict rank-one accuracy over all 3,072 scored prompts per construction in the main experiment, counting baseline and edited prompts and repeated renderings of identical prompts.

Table B1: Query relevance R (nats) by construction and order group in Experiment 1, with 95% history-bootstrap intervals over 96 histories. The live control has one assignment block, so it has no order split.

Model	Construction	Overall R	Aligned R	Reversed R
Qwen3-8B	Live	32.598 [31.258, 33.967]	—	—
Qwen3-8B	Superseded	1.974 [1.840, 2.109]	2.144 [1.995, 2.301]	1.804 [1.660, 1.947]
Qwen3-8B	Other attribute	1.660 [1.488, 1.835]	1.679 [1.527, 1.835]	1.641 [1.422, 1.863]
Qwen3-8B	Entity mention	5.478 [5.128, 5.825]	4.493 [4.235, 4.770]	6.463 [5.971, 6.940]
Qwen3-8B	Early unassigned	−0.037 [−0.203, 0.121]	3.252 [3.046, 3.455]	−3.327 [−3.518, −3.150]
Qwen3-8B	Late unassigned	0.001 [−0.208, 0.211]	3.187 [2.948, 3.442]	−3.184 [−3.433, −2.945]
Gemma 3 4B	Live	39.355 [38.303, 40.415]	—	—
Gemma 3 4B	Superseded	1.688 [1.568, 1.817]	1.703 [1.574, 1.841]	1.672 [1.529, 1.831]
Gemma 3 4B	Other attribute	1.683 [1.541, 1.832]	1.763 [1.598, 1.932]	1.603 [1.453, 1.771]
Gemma 3 4B	Entity mention	4.277 [4.029, 4.545]	5.181 [4.884, 5.496]	3.372 [3.131, 3.633]
Gemma 3 4B	Early unassigned	0.113 [−0.023, 0.245]	2.598 [2.395, 2.798]	−2.372 [−2.567, −2.178]
Gemma 3 4B	Late unassigned	0.125 [−0.055, 0.305]	3.445 [3.202, 3.696]	−3.194 [−3.447, −2.945]

Figure 2: Query relevance R (nats) for the five non-live constructions of Experiment 1, by order group and model. Points are means over 96 histories; intervals are 95% history-bootstrap intervals. Live is omitted from the shared scale (Table B1). Paired order differences are in Table B3.

Restricting the analysis to histories in which every prompt was answered correctly retains 91/96 Qwen and 95/96 Gemma histories and leaves the contrasts essentially unchanged (Table B7).

B.4 Source and donor score changes

Table B8 gives the absolute log masses of the source, donor, and current candidates before and after the edit, and Table B9 the mean changes. The source and donor masses are low in every construction except live, so a change of several nats between them does not imply that probability moves from one to the other.

Table B9 gives the mean baseline-to-edit change in the source and donor scores, averaged over both queries; their difference is the mean edit effect E . Unlike R , E does not subtract the other query’s effect. These are means over prompt pairs without intervals. Because the scores are log masses, exponentiating a mean

Table B2: Superseded minus each control construction, $R_{\text{superseded}} - R_{\text{control}}$ (nats), with 95% history-bootstrap intervals over 96 histories. Because early-unassigned R is near zero after averaging over orders (Section 5.3), the last contrast is close to superseded R itself. Order-stratified versions are in Table B4.

Control	Isolates	Qwen3-8B	Gemma 3 4B
Other attribute	relation status	0.314 [0.147, 0.482]	0.004 [-0.104, 0.118]
Entity mention	construction	-3.504 [-3.806, -3.198]	-2.589 [-2.829, -2.361]
Early unassigned	none: order-averaged control near zero	2.011 [1.855, 2.170]	1.575 [1.440, 1.719]

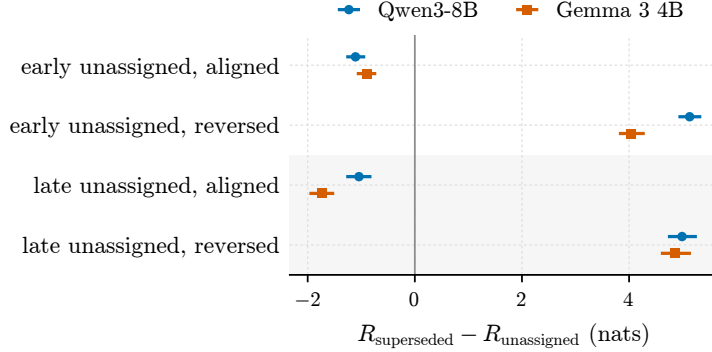


Figure 3: Superseded-minus-unassigned relevance contrasts by order group, in nats, with 95% history-bootstrap intervals. The contrasts are negative under aligned order and positive under reversed order in both models. The comparison depends on both construction and order; it does not imply that construction is irrelevant.

gives a geometric mean of masses. The current answer’s mean log mass rounds to 0.000 before and after editing in every non-live construction except Qwen’s late unassigned (-0.004) and entity mention (-0.001) and Gemma’s early unassigned (-0.004 after editing) and other attribute (-0.001 after editing).

B.5 Target attributes

Each target attribute contributes 24 histories. Superseded minus entity mention is negative for every attribute in both models (Table B10); superseded minus other attribute varies by attribute in Qwen and has intervals spanning zero for every attribute in Gemma (Table B11).

B.6 Cell-stratified bootstrap

The primary bootstrap resamples histories without regard to the $48 \text{ entity-pair} \times \text{attribute} \times \text{name-assignment cells}$. As a sensitivity check we resampled two histories with replacement within each cell on every draw (2,000 draws), which holds the cell counts fixed. Means are unchanged, intervals narrow, and every conclusion about the sign of a contrast is preserved (Table B12).

Table B3: Paired order effect on query relevance, $R_{\text{aligned}} - R_{\text{reversed}}$ (nats), computed within history, with the fraction of the 96 histories in which it is positive.

Construction	Qwen3-8B	Gemma 3 4B
Superseded	0.340 [0.218, 0.470]; 0.688	0.032 [-0.101, 0.168]; 0.531
Entity mention	-1.969 [-2.344, -1.604]; 0.135	1.810 [1.580, 2.058]; 0.958
Early unassigned	6.578 [6.356, 6.789]; 1.000	4.970 [4.700, 5.257]; 1.000
Late unassigned	6.371 [6.113, 6.646]; 1.000	6.639 [6.308, 6.967]; 1.000

Table B4: Superseded minus control, $R_{\text{superseded}} - R_{\text{control}}$ (nats), by order group. Overall values are in Table B2.

Control	Order	Qwen3-8B	Gemma 3 4B
Other attribute	Aligned	0.465 [0.295, 0.649]	-0.060 [-0.215, 0.099]
Other attribute	Reversed	0.162 [-0.049, 0.367]	0.069 [-0.080, 0.213]
Entity mention	Aligned	-2.350 [-2.572, -2.125]	-3.478 [-3.769, -3.197]
Entity mention	Reversed	-4.659 [-5.119, -4.207]	-1.700 [-1.946, -1.465]
Early unassigned	Aligned	-1.108 [-1.281, -0.925]	-0.895 [-1.084, -0.719]
Early unassigned	Reversed	5.130 [4.920, 5.348]	4.044 [3.810, 4.292]
Late unassigned	Overall	1.973 [1.762, 2.183]	1.562 [1.390, 1.742]
Late unassigned	Aligned	-1.043 [-1.281, -0.810]	-1.741 [-1.968, -1.505]
Late unassigned	Reversed	4.988 [4.724, 5.266]	4.866 [4.592, 5.155]

Table B5: Paired aligned-minus-reversed differences in the superseded-minus-control contrast (nats).

Control	Qwen3-8B	Gemma 3 4B
Other attribute	0.302 [0.103, 0.509]	-0.128 [-0.323, 0.081]
Entity mention	2.309 [1.933, 2.707]	-1.778 [-2.049, -1.519]
Early unassigned	-6.238 [-6.481, -5.983]	-4.939 [-5.275, -4.620]
Late unassigned	-6.031 [-6.307, -5.754]	-6.607 [-6.951, -6.243]

Table B6: Strict rank-one candidate accuracy in Experiment 1, per construction.

Construction	Qwen3-8B	Gemma 3 4B
Live	3,064/3,072 (99.7396%)	3,072/3,072 (100%)
Superseded	3,072/3,072 (100%)	3,072/3,072 (100%)
Other attribute	3,072/3,072 (100%)	3,072/3,072 (100%)
Entity mention	3,072/3,072 (100%)	3,072/3,072 (100%)
Early unassigned	3,072/3,072 (100%)	3,071/3,072 (99.9674%)
Late unassigned	3,069/3,072 (99.9023%)	3,072/3,072 (100%)

Table B7: Superseded minus control (nats) among histories with every prompt ranked correctly.

Control	Qwen3-8B (91 histories)	Gemma 3 4B (95 histories)
Other attribute	0.371 [0.201, 0.532]	0.004 [-0.106, 0.119]
Entity mention	-3.445 [-3.750, -3.141]	-2.591 [-2.827, -2.354]
Early unassigned	2.015 [1.861, 2.176]	1.576 [1.433, 1.714]

Table B8: Absolute candidate log masses S (nats) in Experiment 1: mean over prompts, with the first and third quartiles below. Source is the edited earlier value, donor its replacement, current the correct answer; 1,536 baseline and 1,536 edited prompts per construction and model. In the live construction the source is the current value of the edited entity, so its mass is high when that entity is queried.

Construction	Source		Donor		Current	
	Baseline	Edited	Baseline	Edited	Baseline	Edited
<i>Qwen3-8B</i>						
Live	-8.027 (-16.124, -0.000)	-23.650 (-26.373, -20.584)	-24.086 (-26.850, -20.941)	-8.132 (-16.394, -0.000)	-0.004 (-0.000, -0.000)	-0.012 (-0.000, -0.000)
Superseded	-22.271 (-24.703, -19.578)	-23.881 (-26.425, -21.172)	-23.618 (-25.922, -20.888)	-22.120 (-24.476, -19.328)	-0.000 (-0.000, -0.000)	-0.000 (-0.000, -0.000)
Other attribute	-21.048 (-23.317, -18.476)	-22.396 (-24.662, -19.810)	-22.234 (-24.235, -19.717)	-20.903 (-22.978, -18.297)	-0.000 (-0.000, -0.000)	-0.000 (-0.000, -0.000)
Entity mention	-19.836 (-23.242, -16.699)	-23.668 (-26.370, -20.562)	-23.452 (-26.000, -20.390)	-19.868 (-23.344, -16.464)	-0.001 (-0.000, -0.000)	-0.001 (-0.000, -0.000)
Early unassigned	-16.488 (-18.781, -14.193)	-20.697 (-22.808, -18.257)	-20.497 (-22.524, -18.016)	-16.485 (-18.609, -13.922)	-0.000 (-0.000, -0.000)	-0.000 (-0.000, -0.000)
Late unassigned	-17.218 (-19.385, -14.906)	-23.119 (-25.287, -20.589)	-22.899 (-25.107, -20.467)	-17.216 (-19.365, -15.031)	-0.004 (-0.000, -0.000)	-0.004 (-0.000, -0.000)
<i>Gemma 3 4B</i>						
Live	-9.784 (-19.619, -0.000)	-24.049 (-26.585, -21.125)	-24.191 (-27.076, -20.906)	-10.049 (-20.275, -0.000)	-0.000 (-0.000, -0.000)	-0.000 (-0.000, -0.000)
Superseded	-24.311 (-26.603, -22.109)	-25.552 (-27.981, -23.386)	-25.498 (-27.876, -23.237)	-24.268 (-26.601, -21.979)	-0.000 (-0.000, -0.000)	-0.000 (-0.000, -0.000)
Other attribute	-23.486 (-25.900, -21.062)	-24.929 (-27.427, -22.533)	-24.734 (-27.127, -22.564)	-23.621 (-25.952, -21.291)	-0.000 (-0.000, -0.000)	-0.001 (-0.000, -0.000)
Entity mention	-20.540 (-23.446, -17.851)	-23.749 (-26.189, -21.445)	-23.606 (-25.977, -21.137)	-20.650 (-23.591, -17.786)	-0.000 (-0.000, -0.000)	-0.000 (-0.000, -0.000)
Early unassigned	-20.919 (-23.138, -18.691)	-24.235 (-26.460, -22.135)	-23.971 (-26.234, -21.627)	-20.996 (-23.279, -18.750)	-0.000 (-0.000, -0.000)	-0.004 (-0.000, -0.000)
Late unassigned	-20.887 (-23.310, -18.196)	-25.398 (-28.104, -22.651)	-25.138 (-27.712, -22.286)	-21.139 (-23.869, -18.339)	-0.000 (-0.000, -0.000)	-0.000 (-0.000, -0.000)

Table B9: Mean source and donor log-mass changes and mean edit effect E (nats).

Model	Construction	Source change	Donor change	Mean E
Qwen3-8B	Superseded	-1.610	1.497	3.108
Qwen3-8B	Other attribute	-1.348	1.331	2.679
Qwen3-8B	Entity mention	-3.832	3.584	7.417
Qwen3-8B	Early unassigned	-4.208	4.012	8.221
Qwen3-8B	Late unassigned	-5.901	5.682	11.583
Gemma 3 4B	Superseded	-1.241	1.230	2.471
Gemma 3 4B	Other attribute	-1.444	1.113	2.557
Gemma 3 4B	Entity mention	-3.209	2.957	6.166
Gemma 3 4B	Early unassigned	-3.315	2.975	6.290
Gemma 3 4B	Late unassigned	-4.511	3.999	8.510

Table B10: Superseded minus entity mention by target attribute (nats; 24 histories each).

Attribute	Qwen3-8B	Gemma 3 4B
Badge	-4.310 [-4.819, -3.787]	-1.794 [-2.047, -1.556]
Color	-1.794 [-2.129, -1.469]	-2.532 [-2.852, -2.203]
Code	-4.570 [-5.098, -4.058]	-3.937 [-4.418, -3.452]
Label	-3.342 [-3.668, -3.017]	-2.094 [-2.358, -1.809]

Table B11: Superseded minus other attribute by target attribute (nats; 24 histories each).

Attribute	Qwen3-8B	Gemma 3 4B
Badge	0.748 [0.512, 0.969]	0.011 [−0.206, 0.232]
Color	0.876 [0.683, 1.068]	−0.028 [−0.236, 0.208]
Code	−0.570 [−0.866, −0.280]	−0.045 [−0.292, 0.217]
Label	0.200 [−0.042, 0.459]	0.078 [−0.110, 0.264]

Table B12: Superseded minus control (nats) with cell-stratified bootstrap intervals.

Control	Qwen3-8B	Gemma 3 4B
Other attribute	0.314 [0.225, 0.403]	0.004 [−0.079, 0.090]
Entity mention	−3.504 [−3.641, −3.363]	−2.589 [−2.707, −2.475]
Early unassigned	2.011 [1.920, 2.100]	1.575 [1.477, 1.667]

C Provenance of Experiment 1

Models and runtime. Qwen3-8B used checkpoint revision `b968826d9c46dd6066d109eabc6255188de91218` and Gemma 3 4B (`gemma-3-4b-it`) used `093f9f388b31de276ce2de164bdc2081324b9767`, both with int8 weights, CUDA inference, and Python 3.12.13; computation was float16 for Qwen and bfloat16 for Gemma. All scoring used PyTorch 2.14.0 (Paszke et al., 2019) with CUDA 13.0, Hugging Face Transformers 5.17.0 (Wolf et al., 2020), and bitsandbytes 0.50.2 for LLM.int8() weight quantization (Dettmers et al., 2022); the additional models of Appendix F were scored with the same library versions under Python 3.13.5.

Scoring formats. An earlier version of the competence screen scored only the first token of each candidate. It exposed a casing and tokenization problem for Phi-4-mini and ranking errors for Mistral-7B. All results in this paper use the complete-continuation score of Section 3.2 and screens run under it; results from the earlier format are not mixed in.

History freshness. A first confirmation attempt for Gemma stopped at a validation check before the model was loaded and produced no scores. Its 96 histories were excluded from the dataset used here, and a check of history signatures confirms zero overlap. The validator was then corrected; the saved screening scores, re-validated under the corrected implementation, gave the same screening results. Screening and confirmation histories do not overlap with each other or with any earlier stage.

Completion. Each model’s confirmation produced 18,432 valid scored rows over 96 histories with no row errors. The two models were scored on byte-identical datasets, so every comparison between them uses the same histories, values, and donors.

Edit-span geometry. The geometry audits behind the one-token edit spans of Section 4.3 cover all 9,216 baseline/edit pairs per model; their hashes are listed in the supplementary material.

Hashes and the artifact package. The SHA-256 hashes in the supplement identify the exact files behind each reported number; they establish identity, not the time at which a file was recorded. The anonymous artifact package contains the design records, datasets, candidate and tokenizer audits, score files with the rendered prompts, analysis records, model revisions, generation and parser settings, and the dependency lock file, with a manifest of their hashes. Sealed records written on the experiment machines contain home-directory paths and host names; the packaged copies replace these with neutral labels, and the manifest lists each such file’s original hash, as printed here, beside the hash of its packaged copy.

Prespecification record. The decision rule for the original hypothesis was recorded on 3 October 2026, before the confirmation results were examined. It required the lower 95% bound of all three overall superseded-minus-control contrasts (early unassigned, entity mention, other attribute) to exceed zero, and the same for their reversed-order versions to support a claim under reversed order.

D Experiment 2 details

D.1 Provenance

Experiment 2 was run twice with the same design, generator, and scoring code. An earlier run used 24 histories (generation seed 23261011) and excluded a two-history pilot. The run reported in the main text uses 96 histories (generation seed 23261012) and excludes both the pilot and the 24-history run; its design document was hash-stamped on the experiment machine before the dataset was generated. Both runs scored every prompt (24-history run: 3,072 prompts, 2,272 distinct; 96-history run: 12,288 prompts, 8,544 distinct), with scoring seed 20261006, the checkpoint revisions of Appendix C, Python 3.13.5, an NVIDIA GeForce RTX 5080 with CUDA 13.0, and the int8 configurations of Experiment 1. Experiment 2 records candidate scores only, not generated answers. Artifact hashes are listed in the supplementary material.

Phi-4-mini was scored separately on the same 24-history dataset as an exploratory model extension, not as part of the prespecified Qwen/Gemma confirmation. Its pinned revision was

Table D1: The four historical-line templates of Experiment 2, shown for Nora. Each prompt has two such lines, one per entity, followed by the current-state block and query of Experiment 1.

Construction	Marker absent	Marker present
Superseded	Nora’s badge was amber.	Previously, Nora’s badge was amber.
Entity mention	Nora mentioned amber in an unrelated note.	Previously, Nora mentioned amber in an unrelated note.

Table D2: Marker effects, $R_{\text{present}} - R_{\text{absent}}$, and the marker-by-construction interaction (superseded minus entity-mention marker effect) in Experiment 2, in nats, with 95% history-bootstrap intervals over 96 histories. The underlying R values, their source/donor components, and the earlier 24-history run are in Appendix D.

Model	Order	Superseded	Entity mention	Interaction
Qwen3-8B	All	-0.652 [-0.748, -0.560]	-1.734 [-1.934, -1.536]	1.082 [0.850, 1.308]
Qwen3-8B	Aligned	-0.366 [-0.484, -0.243]	-1.000 [-1.164, -0.834]	0.635 [0.427, 0.841]
Qwen3-8B	Reversed	-0.938 [-1.071, -0.804]	-2.469 [-2.746, -2.189]	1.530 [1.208, 1.847]
Gemma 3 4B	All	-0.950 [-1.062, -0.841]	-1.075 [-1.166, -0.986]	0.125 [-0.012, 0.265]
Gemma 3 4B	Aligned	-0.963 [-1.100, -0.834]	-1.445 [-1.566, -1.326]	0.482 [0.307, 0.655]
Gemma 3 4B	Reversed	-0.937 [-1.112, -0.767]	-0.705 [-0.833, -0.580]	-0.232 [-0.436, -0.027]

cfb6facb99257ffa30c83adab238a50856ac3083; it used the RTX 5080 with CUDA 13.0, int8 weights, and float16 activations. Its saved analysis recomputes exactly from all 3,072 score rows. As for the primary models, these are candidate-score results only and do not measure generated-answer accuracy.

D.2 Templates and marker effects

Table D1 gives the four historical-line templates, Table D2 the marker effects and their interaction by order group, and Figure 4 the underlying relevance.

D.3 Query relevance and its components

E can be split into the change in the donor’s score minus the change in the source’s score. Applying the R transformation to each part gives $R = R_{\text{donor}} - R_{\text{source}}$. Table D3 reports R and both components in the 96-history run. With the marker, the donor component is smaller and the source component less negative, at the mean level, in both constructions and both models.

Table D3: Overall R and its donor and source components by construction and marker state in the 96-history run (nats).

Model	Construction	Previously	R	Donor component	Source component
Qwen3-8B	Superseded	Absent	3.486 [3.357, 3.628]	1.622 [1.443, 1.810]	-1.864 [-2.075, -1.654]
Qwen3-8B	Superseded	Present	2.834 [2.721, 2.946]	1.448 [1.274, 1.621]	-1.386 [-1.581, -1.199]
Qwen3-8B	Entity mention	Absent	6.545 [6.250, 6.866]	3.106 [2.849, 3.363]	-3.438 [-3.743, -3.166]
Qwen3-8B	Entity mention	Present	4.810 [4.619, 5.015]	2.281 [2.087, 2.493]	-2.530 [-2.719, -2.345]
Gemma 3 4B	Superseded	Absent	2.502 [2.351, 2.663]	1.244 [1.115, 1.374]	-1.258 [-1.425, -1.097]
Gemma 3 4B	Superseded	Present	1.552 [1.453, 1.654]	0.793 [0.666, 0.925]	-0.759 [-0.893, -0.620]
Gemma 3 4B	Entity mention	Absent	3.409 [3.252, 3.575]	1.633 [1.483, 1.783]	-1.776 [-1.935, -1.619]
Gemma 3 4B	Entity mention	Present	2.334 [2.204, 2.474]	1.104 [0.976, 1.246]	-1.230 [-1.373, -1.080]

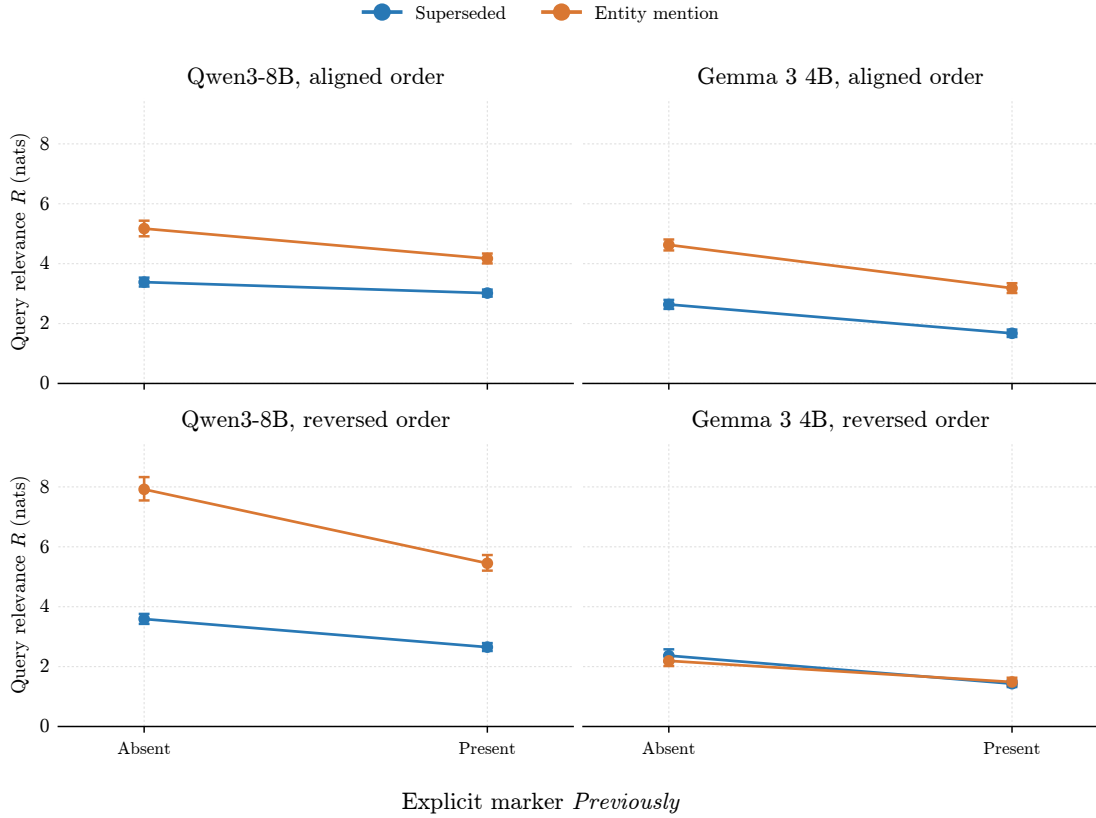


Figure 4: Query relevance in Experiment 2 by marker and construction, separately for aligned (top) and reversed (bottom) order. Points are means over 96 histories; intervals are marginal 95% history-bootstrap intervals. In Gemma the lines are further apart without the marker under aligned order and converge under reversed order, so the marker-by-construction interaction changes sign between order groups; paired marker effects and interactions are in Table D2.

D.4 The earlier 24-history run

The 24-history run agrees with the 96-history run in part (Table D4): every marker effect is negative, Qwen’s pooled interaction is positive, and Gemma’s pooled interaction interval includes zero. Gemma’s order-specific interaction intervals also include zero, so this run does not show the order crossover found in the 96-history run. Its R decomposition and paired component changes are in Tables D5 and D6. The two runs are not pooled.

Table D4: Marker effects and marker-by-construction interaction in the 24-history run (nats; 95% history-bootstrap intervals).

Model	Order	Superseded	Entity mention	Interaction
Qwen3-8B	All	−0.750 [−0.917, −0.590]	−2.087 [−2.410, −1.772]	1.337 [0.976, 1.712]
Qwen3-8B	Aligned	−0.372 [−0.552, −0.187]	−1.298 [−1.580, −0.979]	0.926 [0.581, 1.274]
Qwen3-8B	Reversed	−1.127 [−1.324, −0.939]	−2.876 [−3.354, −2.429]	1.749 [1.236, 2.274]
Gemma 3 4B	All	−1.085 [−1.308, −0.864]	−0.960 [−1.183, −0.727]	−0.125 [−0.390, 0.136]
Gemma 3 4B	Aligned	−1.100 [−1.359, −0.851]	−1.155 [−1.384, −0.915]	0.055 [−0.233, 0.328]
Gemma 3 4B	Reversed	−1.069 [−1.372, −0.785]	−0.765 [−1.123, −0.387]	−0.304 [−0.753, 0.116]

Table D5: Overall R and its donor and source components by construction and marker state in the 24-history run (nats).

Model	Construction	Previously	R	Donor component	Source component
Qwen3-8B	Superseded	Absent	3.738 [3.536, 3.934]	1.590 [1.118, 2.005]	-2.149 [-2.623, -1.705]
Qwen3-8B	Superseded	Present	2.989 [2.791, 3.186]	1.283 [0.832, 1.746]	-1.705 [-2.151, -1.242]
Qwen3-8B	Entity mention	Absent	7.304 [6.629, 7.966]	3.875 [3.477, 4.277]	-3.429 [-4.135, -2.764]
Qwen3-8B	Entity mention	Present	5.217 [4.749, 5.705]	2.540 [2.171, 2.917]	-2.677 [-3.193, -2.165]
Gemma 3 4B	Superseded	Absent	2.631 [2.409, 2.863]	1.413 [1.170, 1.652]	-1.218 [-1.501, -0.934]
Gemma 3 4B	Superseded	Present	1.546 [1.379, 1.714]	0.685 [0.445, 0.932]	-0.861 [-1.115, -0.610]
Gemma 3 4B	Entity mention	Absent	3.313 [2.975, 3.669]	1.732 [1.427, 2.048]	-1.582 [-1.915, -1.276]
Gemma 3 4B	Entity mention	Present	2.353 [2.114, 2.605]	1.079 [0.874, 1.287]	-1.274 [-1.468, -1.068]

Table D6: Paired marker-induced changes (present minus absent) in the donor and source components in the 24-history run (nats); $\Delta R = \Delta R_{\text{donor}} - \Delta R_{\text{source}}$.

Model	Order	Construction	Δ donor component	Δ source component
Qwen3-8B	All	Superseded	-0.306 [-0.812, 0.237]	0.443 [-0.100, 1.040]
Qwen3-8B	All	Entity mention	-1.335 [-1.640, -1.013]	0.752 [0.323, 1.140]
Qwen3-8B	Aligned	Superseded	-0.286 [-0.840, 0.267]	0.086 [-0.522, 0.704]
Qwen3-8B	Aligned	Entity mention	-0.810 [-1.328, -0.330]	0.488 [-0.179, 1.070]
Qwen3-8B	Reversed	Superseded	-0.327 [-0.995, 0.399]	0.800 [0.088, 1.557]
Qwen3-8B	Reversed	Entity mention	-1.859 [-2.283, -1.435]	1.017 [0.480, 1.492]
Gemma 3 4B	All	Superseded	-0.727 [-1.046, -0.420]	0.357 [-0.009, 0.720]
Gemma 3 4B	All	Entity mention	-0.653 [-1.003, -0.310]	0.307 [-0.021, 0.645]
Gemma 3 4B	Aligned	Superseded	-0.547 [-1.049, -0.015]	0.553 [0.062, 1.085]
Gemma 3 4B	Aligned	Entity mention	-0.679 [-1.101, -0.245]	0.475 [0.056, 0.909]
Gemma 3 4B	Reversed	Superseded	-0.907 [-1.298, -0.486]	0.162 [-0.301, 0.629]
Gemma 3 4B	Reversed	Entity mention	-0.626 [-1.085, -0.207]	0.139 [-0.285, 0.532]

D.5 Exploratory Phi-4-mini extension

For Phi-4-mini, adding the explicit prefix **Previously**, changed R in the superseded construction by -1.355 nats (95% CI $[-1.561, -1.138]$), from 5.146 $[4.947, 5.362]$ without the marker to 3.791 $[3.592, 4.004]$ with it. In the entity-mention construction, the marker effect was 0.145 $[-0.018, 0.310]$, an unresolved change from 4.808 $[4.557, 5.107]$ to 4.953 $[4.677, 5.269]$. The paired marker-by-construction interaction was -1.500 $[-1.734, -1.264]$, indicating a larger reduction for the superseded construction in this model.

The order-stratified estimates show additional heterogeneity. Superseded marker effects were -1.227 $[-1.412, -1.026]$ in aligned order and -1.483 $[-1.758, -1.194]$ in reversed order. Entity-mention effects were -0.253 $[-0.450, -0.059]$ aligned and 0.544 $[0.384, 0.703]$ reversed. The corresponding interaction was -0.973 $[-1.221, -0.714]$ aligned and -2.027 $[-2.330, -1.725]$ reversed. These exploratory estimates indicate construction- and order-dependent score effects for Phi-4-mini; they do not establish a general model-family pattern or identify a mechanism. They are effects of adding the literal prefix, not evidence of active suppression. No answer-generation accuracy was collected in this candidate-scoring run.

Table D7: Exploratory Phi-4-mini marker-confirmation estimates (nats; 24 histories; 95% history-bootstrap intervals). Marker effect is present minus absent; interaction is the superseded marker effect minus the entity-mention marker effect.

Order	Superseded marker effect	Entity-mention marker effect	Interaction
All	-1.355 [-1.561, -1.138]	0.145 [-0.018, 0.310]	-1.500 [-1.734, -1.264]
Aligned	-1.227 [-1.412, -1.026]	-0.253 [-0.450, -0.059]	-0.973 [-1.221, -0.714]
Reversed	-1.483 [-1.758, -1.194]	0.544 [0.384, 0.703]	-2.027 [-2.330, -1.725]

E An attempt to lower accuracy with distractors

Because both models answer at ceiling, we tried to make the task harder by adding distractor entities to the Experiment 1 constructions. Each distractor contributed its own **Previously** line and **Currently** line for the target attribute, with values from a separate vocabulary. The plan was to choose, on development data, the hardest of three levels (2, 4, or 6 distractors) at which Qwen’s accuracy on generated answers fell between 65% and 90%, and then to test whether edits to the earlier value change generated answers. Before the development results were available, we added a stopping rule: if Qwen exceeded 90% at all three levels, the study would stop without a test.

Each development level used 12 matched target histories and 1,536 scored prompt members (1,152 unique prompts) per model, with greedy decoding and exact matching of the answer. Prompt members include repeated renderings across paired cells; the three levels are matched development conditions, not independent replications. Qwen exceeded 90% at every level (Table E1), so the prespecified stopping rule ended the study: no level was selected and no test data were generated. Gemma was also near ceiling and is not an informative behavioral replication. Qwen’s few source-value responses occurred in reversed-order entity-mention controls, where the queried attribute had not been assigned that value; they are not stale-assignment errors. The superseded construction made up a quarter of the prompts (384 per level and model, 1,152 across the three levels), and Qwen and Gemma answered every one of them with the current value, never with the earlier one.

Phi-4-mini was also scored on the same three development datasets as an additional, exploratory model; it was not part of the prespecified level-selection rule. Its exact-match accuracy was 8/1,536 (0.52%), 38/1,536 (2.47%), and 66/1,536 (4.30%) at 2, 4, and 6 distractors. These rates are dominated by response-format failures: in a post hoc string-prefix diagnostic, respectively 1,516, 1,530, and 1,527 outputs began with the expected candidate label, but continued with extra text and therefore remained incorrect under the exact-answer parser. The prefix diagnostic is descriptive and does not redefine correctness. Phi’s strict accuracy cannot be interpreted as a response to increasing distractor difficulty, and its floor-level exact-match outcomes do not provide an informative behavioral replication. No Phi test was generated.

This was a limited difficulty manipulation. Distractors used a separate vocabulary and were appended after the target blocks; each distractor’s historical and current assignment lines were adjacent. The results show that this specific manipulation did not create the intended difficulty for Qwen. They do not show that the results hold in long, crowded, or conflicting contexts in general.

Table E1: Strict exact-match accuracy of generated answers in the distractor development runs, by number of distractor entities.

Model	2 distractors	4 distractors	6 distractors
Qwen3-8B	1,531/1,536 (99.67%)	1,532/1,536 (99.74%)	1,533/1,536 (99.80%)
Gemma 3 4B	1,534/1,536 (99.87%)	1,532/1,536 (99.74%)	1,532/1,536 (99.74%)
Phi-4-mini	8/1,536 (0.52%)	38/1,536 (2.47%)	66/1,536 (4.30%)

Table F1: Name-value distance follow-up: R (nats) by construction and distance, near-minus-far differences, and the entity-minus-superseded gap at each distance, with 95% history-bootstrap intervals over 96 histories. Measured name-value gaps: one token in both near conditions; three (Qwen) or four (Gemma) tokens for superseded far; seven for entity-mention far.

	Qwen3-8B	Gemma 3 4B
Superseded, near	2.685 [2.516, 2.861]	2.005 [1.858, 2.135]
Superseded, far	2.184 [2.030, 2.335]	1.612 [1.492, 1.735]
Entity mention, near	3.977 [3.684, 4.264]	2.372 [2.245, 2.505]
Entity mention, far	4.347 [4.043, 4.662]	2.399 [2.261, 2.545]
Superseded, near – far	0.500 [0.400, 0.599]	0.393 [0.288, 0.496]
Entity mention, near – far	−0.370 [−0.498, −0.249]	−0.027 [−0.110, 0.058]
Entity – superseded, near	1.293 [1.019, 1.571]	0.367 [0.234, 0.514]
Entity – superseded, far	2.163 [1.866, 2.459]	0.788 [0.657, 0.922]

F Follow-up experiment details

The follow-ups of Section 4.3 were run on the RTX 5080 used for the main experiments, with the same pinned checkpoints, int8 configurations, and scoring code. Each design document was hash-stamped on the experiment machine before its data were generated (not committed to version control; Section 4.2). Artifact hashes are listed in the supplementary material.

F.1 Name-value distance

Design. Entity mentions place the value one word after the name (Nora mentioned amber), superseded assignments several tokens later (Nora’s badge was amber). We crossed construction (superseded, entity mention) with name-value distance (near, far), all lines marked with **Previously**: superseded far **Previously, Nora’s badge was amber.**, superseded near **Previously, the badge of Nora was amber.**, entity-mention near **Previously, Nora mentioned amber in an unrelated note.**, and entity-mention far **Previously, Nora, in an unrelated note, mentioned amber.** A tokenizer audit, run before data generation, measured the token gap between name and value: one token in both near conditions for both models, three (Qwen) or four (Gemma) for superseded far, and seven for entity-mention far. The first near superseded wordings we considered (Nora’s badge: amber and two variants) did not shorten the measured gap and were replaced before any data were generated (Appendix F). The study uses 96 histories with Experiment 1’s allocation and 12,288 scored prompts per model. The recorded prediction was that near exceeds far within each construction, with lower 95% bounds above zero in both constructions.

Choice of the near superseded template. The design listed three near superseded wordings (Nora’s badge: amber, Nora’s badge = amber, Nora’s badge, amber) and a rule choosing the first whose base-line/edit pairs differ only in the edited value’s tokens for both tokenizers. A tokenizer audit run before data generation found that all three pass that check but leave the measured name-value gap unchanged (three tokens in Qwen, four in Gemma, as for the far template), so the superseded arm would not have varied distance. Before any data were generated, an amendment added a fourth candidate, **Previously, the badge of Nora was amber.**, and required the chosen template to be measurably nearer than the far template in both tokenizers; the re-run audit chose it, with a gap of one token in both. The amendment was hash-stamped separately.

Data. 96 histories (Experiment 1’s allocation of entity pairs, attributes, and name orientations), excluding all histories of the marker pilot and the 24-history marker run; 12,288 scored prompts per model (9,152 distinct); scoring took 6,472 s (Qwen) and 2,656 s (Gemma). Results are in Table F1; the distance-by-construction interaction (superseded near-minus-far minus entity-mention near-minus-far) was 0.871 [0.703, 1.040] in Qwen and 0.420 [0.291, 0.542] in Gemma.

F.2 Update chains

Design. Chain depth is the total number of successive assignments of the queried attribute, including the current one, so depth 3 means two updates. To test generated answers below ceiling, each entity received a chain of three, four, or five assignments revealed in rounds (**Initially, ...was v., Then, ...changed to v., Currently, ...is v.**); the edit replaced the immediately superseded value, and the vocabulary was extended to twelve values chosen by a recorded tokenizer audit. Models generated answers greedily and were also scored on candidate masses. A development run of 12 histories per depth was to select the shallowest depth at which at least one model’s answer accuracy fell between 70% and 95%, and a 96-history confirmation would then measure whether edits move generated answers toward the donor value; if no depth qualified, the study would stop. Because no depth qualified, a second run scored candidate masses only on 96 fresh histories per depth, to estimate how R changes with depth; the recorded rule counted growth as supported if R at depth 5 minus R at depth 3 had a lower 95% bound above zero, comparing independent history sets. Development: 384 generated answers per depth and model.

Vocabulary. Disjoint chains of up to five values per entity plus two donors need twelve values. The design extended the eight-value vocabulary by walking a recorded candidate list in order and keeping each value that is not a substring of another and keeps the candidate continuations valid for both tokenizers; the audit kept the first four candidates, **olive, plum, navy, and rust**.

Development. 12 histories per depth (32 prompts per history: historical order $\times$ current order $\times$ edited entity $\times$ query $\times$ baseline/edit), greedy decoding with at most 32 new tokens, and the exact-answer parser of Appendix E. Table F2 reports accuracy, the share of answers naming any earlier value, and R from candidate masses over the twelve values. Every answer was correct for both models at every depth, so the recorded selection rule chose no depth and the confirmation was not generated.

Table F2: Update-chain development runs (12 histories, 384 generated answers per depth and model). Accuracy is exact-match accuracy of generated answers; “earlier value” is the share of answers naming any superseded value; R is mean query relevance from candidate masses.

Model	Depth	Accuracy	Earlier value	R (nats)
Qwen3-8B	3	384/384	0	7.940
Qwen3-8B	4	384/384	0	10.612
Qwen3-8B	5	384/384	0	12.040
Gemma 3 4B	3	384/384	0	3.345
Gemma 3 4B	4	384/384	0	3.289
Gemma 3 4B	5	384/384	0	3.674

Candidate-score run. Because no depth qualified for a generation confirmation, a second run scored candidate masses only on 96 fresh histories per depth (dataset seed 20261110; development histories excluded). Table F3 gives R by depth and the depth differences, compared across independent history sets by resampling each set separately.

F.3 Updated other attribute

Design. To separate the two properties joined in superseded versus other attribute, three constructions share a six-line layout: two historical lines, two extra current lines, and the two current lines for the queried attribute. In the superseded construction the historical lines assign the queried attribute and the extra lines assign a filler attribute (**Previously, Nora’s badge was amber.; Currently, Nora’s project is slate.**). In the updated other-attribute construction the historical lines assign another attribute and the extra lines update it (**Previously, Nora’s team was amber.; Currently, Nora’s team is slate.**); in the static other-attribute construction the extra lines assign the filler attribute instead. Superseded minus updated other attribute then measures having been the queried attribute, with both values contradicted,

Table F3: Update chains, candidate-score run (96 histories per depth). R in nats with 95% history-bootstrap intervals; depth differences use independent-set bootstrap intervals.

	Qwen3-8B	Gemma 3 4B
R , depth 3	7.353 [6.972, 7.725]	3.450 [3.229, 3.676]
R , depth 4	10.762 [10.303, 11.220]	3.462 [3.258, 3.667]
R , depth 5	11.835 [11.424, 12.286]	3.708 [3.479, 3.936]
Depth 4 – depth 3	3.410 [2.835, 3.969]	0.012 [−0.297, 0.305]
Depth 5 – depth 4	1.072 [0.469, 1.680]	0.246 [−0.050, 0.552]
Depth 5 – depth 3	4.482 [3.915, 5.047]	0.258 [−0.063, 0.557]

Table F4: Updated-other-attribute follow-up: R (nats) by construction and the two contrasts it separates, with 95% history-bootstrap intervals over 96 histories. All three constructions share a six-line layout.

	Qwen3-8B	Gemma 3 4B
Superseded	1.368 [1.282, 1.457]	0.817 [0.713, 0.928]
Other attribute, updated	0.628 [0.555, 0.700]	0.669 [0.568, 0.770]
Other attribute, not updated	0.917 [0.835, 1.003]	0.697 [0.602, 0.802]
Queried attribute (superseded – updated)	0.740 [0.661, 0.818]	0.148 [0.038, 0.261]
Update (updated – not updated)	−0.289 [−0.363, −0.215]	−0.028 [−0.137, 0.076]
Superseded – not updated	0.451 [0.376, 0.525]	0.120 [0.027, 0.209]

and updated minus static other attribute measures being contradicted by an update, with neither value the queried attribute. The recorded prediction of the discount account was an update effect below zero. The study uses 96 histories with Experiment 1’s allocation, all eight vocabulary values per history, and 9,216 scored prompts per model.

The 96 histories follow Experiment 1’s allocation; the other attribute and the filler attribute are **team** and **project**, swapped between the two halves of the histories, and each history uses all eight vocabulary values (two historical, two extra current, two current for the queried attribute, two donors). Historical order, current order, edited entity, query, and baseline/edit are crossed as in Experiment 1, with the extra current block in the current order; 9,216 prompts per model (6,912 distinct). By order group, the queried-attribute effect was 0.801 [0.691, 0.919] (aligned) and 0.679 [0.579, 0.780] (reversed) in Qwen and 0.362 [0.205, 0.528] and −0.066 [−0.198, 0.064] in Gemma; the update effect was −0.225 [−0.334, −0.117] and −0.354 [−0.452, −0.249] in Qwen and −0.036 [−0.186, 0.099] and −0.019 [−0.150, 0.118] in Gemma.

Correctly ranked histories. Gemma ranks the current candidate first on 2,895 of 3,072 superseded prompts in this design (Table F10). Restricting both models to histories in which every prompt is ranked correctly, as in Table B7, leaves 87 histories for Qwen and 28 for Gemma. Superseded minus updated other attribute is then 0.734 [0.662, 0.811] in Qwen and 0.277 [0.080, 0.481] in Gemma; updated minus static other attribute is −0.276 [−0.353, −0.200] and 0.047 [−0.123, 0.221]; superseded minus static other attribute is 0.458 [0.384, 0.534] and 0.324 [0.201, 0.449]. The queried-attribute effects remain positive and resolved, Qwen’s update effect remains negative and resolved, and Gemma’s update effect remains unresolved. Gemma’s restricted estimates rest on fewer than a third of its histories.

F.4 Generated answers below ceiling (exploratory)

Design. Because Qwen and Gemma answer at ceiling, generated answers were studied in OLMo-2-7B-Instruct, which fails the competence screen (Section 5.9) and so lies outside the screened population. It generated answers greedily (at most 32 tokens, exact match to the eight values) on all 18,432 Experiment 1 confirmation prompts. Generated answers are summarized by *donor following*, a separate descriptive measure

Table F5: Exploratory generated-answer study in OLMo-2-7B-Instruct (outside the screened population) on the Experiment 1 confirmation set. Accuracy counts all 3,072 prompts per construction; donor following is the share of edited prompts answered with the donor under the edited entity’s query minus under the other entity’s query, with 95% history-bootstrap intervals over 96 histories.

Construction	Accuracy	Donor following	Aligned	Reversed
Live	2,970/3,072	0.948 [0.924, 0.969]	0.948	0.948
Superseded	2,950/3,072	0.008 [0.003, 0.014]	0.016	0.000
Other attribute	2,990/3,072	0.001 [0.000, 0.004]	0.000	0.003
Entity mention	1,846/3,072	0.155 [0.124, 0.190]	0.185	0.125
Early unassigned	1,528/3,072	−0.001 [−0.022, 0.020]	0.161	−0.164
Late unassigned	2,131/3,072	0.005 [−0.022, 0.031]	0.148	−0.138

computed on edited prompts only, without baseline adjustment: the share of edited prompts answered with the donor when the edited entity is queried, minus that share when the other entity is queried, computed per history. This study is exploratory and does not enter any confirmatory claim.

Error breakdown and score-based relevance. Table F6 classifies every generated answer. Errors split mainly between the two entities’ earlier values, in about equal numbers, and answers outside the eight-value format; the edited entity’s earlier value (source) and its replacement (donor) appear in the baseline and edited prompts respectively. Table F7 sets OLMo-2’s score-based R , computed from the same run’s candidate masses, beside a baseline-adjusted generation contrast: for each prompt pair, the change in whether the answer is the donor minus the change in whether it is the source, combined across edited entities and queries exactly as E is combined into R . Unlike donor following, this contrast uses the baseline prompts.

Table F6: OLMo-2-7B-Instruct generated answers on the Experiment 1 confirmation set, by construction and prompt version (1,536 prompts each): correct, the edited entity’s earlier value (source), its replacement (donor), the other entity’s current or earlier value, another of the eight values, or no valid value.

Construction	Prompt	Correct	Source	Donor	Other’s current	Other’s earlier	Other value	Invalid
Live	baseline	1,492	6	0	6	0	0	32
	edited	1,478	0	4	16	0	0	38
Superseded	baseline	1,476	7	0	0	7	0	46
	edited	1,474	0	6	1	4	1	50
Other attribute	baseline	1,492	1	0	8	1	2	32
	edited	1,498	0	1	7	1	0	29
Entity mention	baseline	916	145	3	26	145	5	296
	edited	930	1	135	25	146	4	295
Early unassigned	baseline	754	146	0	0	146	0	490
	edited	774	0	137	0	149	0	476
Late unassigned	baseline	1,070	177	0	2	177	0	110
	edited	1,061	0	148	4	209	0	114

F.5 Additional models

Procedure. Each additional model was scored on Experiment 1’s sealed screening dataset (24 histories) and, if it passed, on the sealed confirmation dataset (96 histories), with the same renderer, candidate validation, scorer, competence rule, and analysis code as Qwen and Gemma. All additional models were loaded with int8 weights at pinned revisions. Llama-3.1-8B-Instruct, the planned model from a third family, could not be used because access to its gated weights was declined; Falcon3-7B-Instruct and Granite-3.1-8B-Instruct were added as models from other families, Qwen2.5-7B-Instruct and Gemma 3 12B as other

Table F7: OLMo-2-7B-Instruct: score-based query relevance R (nats) from candidate masses, and the baseline-adjusted generation contrast (change in $P(\text{donor})$ minus change in $P(\text{source})$, query-specific), overall and by order group. Means with 95% history-bootstrap intervals over 96 histories; order-group values are means.

Construction	Score-based R	Generation contrast		
		Overall	Aligned	Reversed
Live	16.864 [16.049, 17.652]	1.911 [1.867, 1.951]	1.911	1.911
Superseded	6.157 [5.933, 6.393]	0.017 [0.007, 0.029]	0.031	0.003
Other attribute	3.744 [3.587, 3.894]	0.003 [0.000, 0.007]	0.000	0.005
Entity mention	5.350 [5.040, 5.669]	0.320 [0.268, 0.379]	0.393	0.247
Early unassigned	0.050 [−0.103, 0.201]	−0.001 [−0.038, 0.035]	0.315	−0.318
Late unassigned	0.061 [−0.125, 0.251]	0.014 [−0.027, 0.061]	0.310	−0.281

generations and sizes within the two original families. Qwen3-14B did not fit in 16 GB of GPU memory and was run on an 80 GB GPU; Qwen2.5-7B produced non-finite logits with float16 computation and was run with bfloat16 computation, before any of its scores were written. Granite’s chat template inserts the current date into every prompt; the rendered prompts are stored with its scores.

Screens. Table F8 gives strict rank-one accuracy on distinct screening prompts. OLMo-2-7B-Instruct (revision 470b1fba1ae01581f270116362ee4aa1b97f4c84, bfloat16 computation) failed in four constructions. Phi-4-mini, re-scored with this procedure, reproduced its original screening result exactly. Falcon3-7B, Qwen3-14B, Gemma 3 12B, Granite-3.1-8B and Mistral-7B-Instruct-v0.3 passed. Mistral had failed an earlier, differently formatted version of the screen (97.57% in one construction) and was screened on Experiment 1 only for the threshold check below; it passed, so by the rule recorded beforehand it is a screened model like the others. Qwen2.5-7B-Instruct failed in late unassigned (563/576).

Table F8: Competence screens of the additional models: correct current value strictly ranked first, over distinct screening prompts (requirement: 99% in every construction).

Model	Live	Superseded	Other attr.	Entity mention	Early unass.	Late unass.	Screen
OLMo-2-7B	287/288	563/576	575/576	426/576	452/576	448/576	fail
Phi-4-mini	288/288	576/576	573/576	561/576	576/576	576/576	fail
Qwen2.5-7B	288/288	576/576	575/576	576/576	576/576	563/576	fail
Falcon3-7B	288/288	576/576	576/576	574/576	576/576	576/576	pass
Granite-3.1-8B	288/288	576/576	576/576	576/576	576/576	576/576	pass
Mistral-7B	288/288	574/576	572/576	571/576	576/576	576/576	pass
Gemma 3 12B	288/288	576/576	576/576	576/576	576/576	576/576	pass
Qwen3-14B	288/288	576/576	576/576	576/576	576/576	576/576	pass

Mistral-7B in detail. Mistral-7B’s entity-mention gap is unresolved overall (−0.415 [−1.000, 0.187]) because it changes sign with mention order: superseded relevance exceeds entity-mention relevance by 2.417 [1.596, 3.224] under aligned order, and entity-mention relevance exceeds superseded relevance by 1.586 [0.969, 2.242] under reversed order. Early-unassigned relevance is 5.975 under aligned and −5.862 under reversed order. Restricted, as in Table B7, to the 68 histories in which every prompt is ranked correctly, the relation-status contrast is −0.037 [−0.680, 0.541], the entity-mention gap −0.447 [−1.039, 0.185], and early-unassigned relevance 5.359 aligned and −4.980 reversed. Mistral-7B had failed an earlier, differently formatted version of the screen and passes this one.

Models excluded by the screen, scored descriptively. Two models failed the screen narrowly, each in a single construction: Phi-4-mini (561/576 in entity mention) and Qwen2.5-7B-Instruct (563/576 in late unassigned). To show whether the conclusions depend on the 99% threshold, each was scored on all 18,432

confirmation prompts with the unchanged scorer and analysis. Two criteria were recorded before scoring: the entity-mention gap (b) exceeds the relation-status contrast (a) in absolute value, which we evaluate as both $(b) - (a)$ and $(b) + (a)$ having lower 95% bounds above zero, and early-unassigned relevance is positive under aligned and negative under reversed order, with both intervals excluding zero. Table F9 gives the results; they are descriptive and enter no confirmatory claim.

Table F9: Experiment 1 in models that failed the competence screen, scored descriptively, in the format of Table 2, with $(b) + (a)$ added. The recorded criterion $(b) > |(a)|$ holds if and only if the lower bounds of both $(b) - (a)$ and $(b) + (a)$ exceed zero. Means with 95% history-bootstrap intervals over 96 histories, in nats.

Model	(a) Superseded – other attribute	(b) Entity mention – superseded	Early unassigned		(b) – (a)	(b) + (a)
			Aligned	Reversed		
Phi-4-mini	−0.152 [−0.266, −0.040]	1.872 [1.664, 2.091]	3.007 [2.871, 3.149]	−2.953 [−3.075, −2.818]	2.025 [1.759, 2.306]	1.720 [1.527, 1.911]
Qwen2.5-7B	−0.553 [−0.736, −0.388]	2.253 [1.931, 2.595]	5.912 [5.605, 6.253]	−5.845 [−6.181, −5.517]	2.806 [2.432, 3.227]	1.700 [1.357, 2.043]

F.6 Candidate-ranking accuracy

Table F10 gives the share of scored prompts, baseline and edited, in which the current value’s candidate mass is strictly higher than every other candidate’s (ties count as errors), for every model and experiment. This is candidate-ranking accuracy, not generated-answer accuracy; the only generated answers are those of Appendix E, the update-chain development run, and OLMo-2.

Table F10: Current-candidate rank accuracy over all scored prompts, with the construction where it is lowest.

Model	Experiment	Correct	Lowest construction
Qwen3-8B	Experiment 1	18,421/18,432	Live 3,064/3,072
Gemma 3 4B	Experiment 1	18,431/18,432	Early unassigned 3,071/3,072
Falcon3-7B	Experiment 1	18,427/18,432	Entity mention 3,069/3,072
Granite-3.1-8B	Experiment 1	18,432/18,432	Early unassigned 3,072/3,072
Mistral-7B	Experiment 1	18,350/18,432	Other attribute 3,020/3,072
Gemma 3 12B	Experiment 1	18,432/18,432	Early unassigned 3,072/3,072
Qwen3-14B	Experiment 1	18,432/18,432	Early unassigned 3,072/3,072
Phi-4-mini (excluded)	Experiment 1	18,316/18,432	Entity mention 2,976/3,072
Qwen2.5-7B (excluded)	Experiment 1	18,381/18,432	Late unassigned 3,040/3,072
OLMo-2-7B (excluded)	Experiment 1	15,687/18,432	Entity mention 2,096/3,072
Qwen3-8B	Experiment 2 (marker)	12,288/12,288	superseded 6,144/6,144
Qwen3-8B	Distance	12,288/12,288	superseded 6,144/6,144
Qwen3-8B	Updated other attribute	9,183/9,216	other updated 3,054/3,072
Qwen3-8B	Chains, depth 3, score run	3,072/3,072	superseded chain 3,072/3,072
Qwen3-8B	Chains, depth 4, score run	3,072/3,072	superseded chain 3,072/3,072
Qwen3-8B	Chains, depth 5, score run	3,072/3,072	superseded chain 3,072/3,072
Gemma 3 4B	Experiment 2 (marker)	12,273/12,288	superseded 6,129/6,144
Gemma 3 4B	Distance	12,288/12,288	superseded 6,144/6,144
Gemma 3 4B	Updated other attribute	8,852/9,216	superseded 2,895/3,072
Gemma 3 4B	Chains, depth 3, score run	3,071/3,072	superseded chain 3,071/3,072
Gemma 3 4B	Chains, depth 4, score run	3,067/3,072	superseded chain 3,067/3,072
Gemma 3 4B	Chains, depth 5, score run	3,070/3,072	superseded chain 3,070/3,072